



# Samuel Bruin

Software Developer

📍 Lelystad, Flevoland  
📞 +31 6 23632314  
✉ [cv@samuelbruin.com](mailto:cv@samuelbruin.com)  
in [linkedin.com/in/samuelbruin](https://linkedin.com/in/samuelbruin)  
🐙 [github.com/ancientkingg](https://github.com/ancientkingg)

[samuelbruin.com](https://samuelbruin.com)

## Work experience

### Head Teaching Assistant @ TU Delft

Sep 2025 – present

*FlexDelft, Delft, The Netherlands*

As Head TA for Web & Databases, Information and Data Management, and Databases & Software Engineering, I coordinated course delivery and administration, curated and maintained course materials, supervised and supported TAs, and guided students by providing structured feedback to support their progress in the course.

### Student Research Developer @ TU Delft

Oct 2024 – June 2025

*TU Delft Algorithmics Group, Delft, The Netherlands*

Developed the Home & Hybrid Energy Management System (HHEMS), focusing on the design and implementation of optimization and control strategies for residential energy management with distributed generation and storage. Worked on system modeling, integration of forecasting components, and evaluation of energy efficiency and cost performance using real-world data.

### Software Engineer Intern

April 2024 – July 2024

*PwC Nederland, Amsterdam, The Netherlands*

As part of this internship, I, along with a team of 5 students, developed an AI-powered, currently still confidential, tool that is now being prepared to go into production for internal use.

### Teaching Assistant for various courses @ TU Delft

Nov 2023 – June 2024

*FlexDelft, Delft, The Netherlands*

As a TA, for various courses such as Calculus, Concepts of Programming Languages, Collaborative Software Engineering Project, Software Project, and Software Engineering Methods, I assisted in course instruction and administration, guided students and offered feedback to help them develop their skills in the course.

## Education

### Master of Science: Computer and Embedded Systems Engineering

Sep 2025 – Present

*Delft University of Technology*

### Bachelor of Science: Computer Science & Engineering (Cum Laude)

Sep 2022 – Jun 2025

*Delft University of Technology (GPA 8.3)*

### Secondary education: VWO

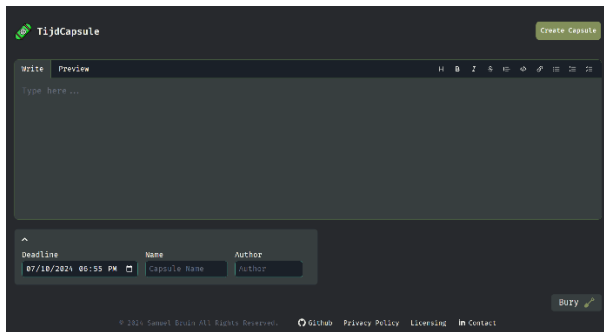
Sep 2016 – Jul 2022

*Lyceum Porteum*

## Portfolio

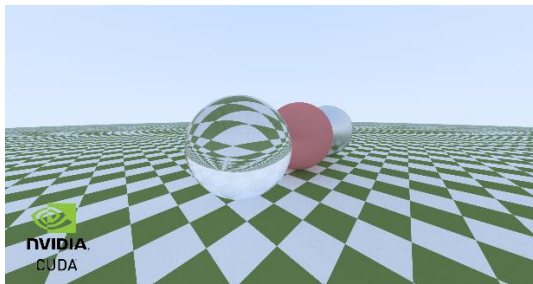
### TijdCapsule – (Svelte, Rust) [<https://tijdcapsule.samuelbruin.com>]

A pastebin alternative with the option of setting a time lock on the content before being able to view it.



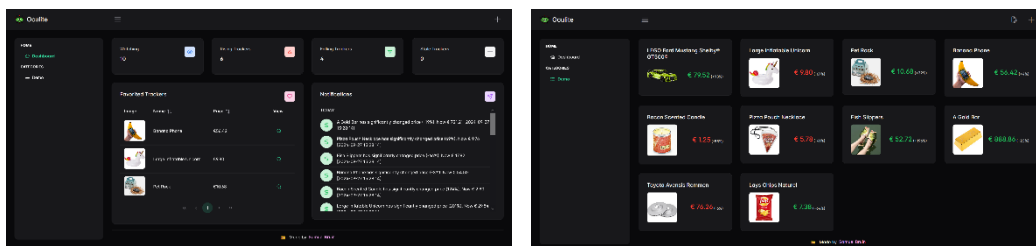
### CUDA Hardware Ray Tracer – (C++, CMAKE, CUDA, OpenGL)

A real-time hardware-accelerated ray tracer written in CUDA displayed by OpenGL, using CUDA and OpenGL interop.



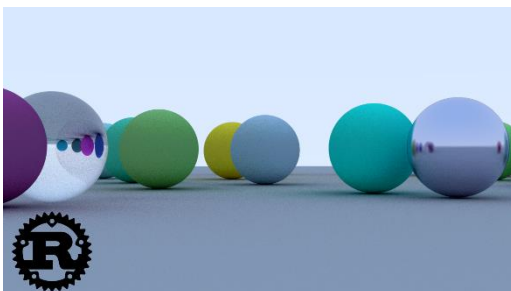
### Oculite – (Rust, VueJS)

A user-friendly dashboard for monitoring prices, written in Vue 3.



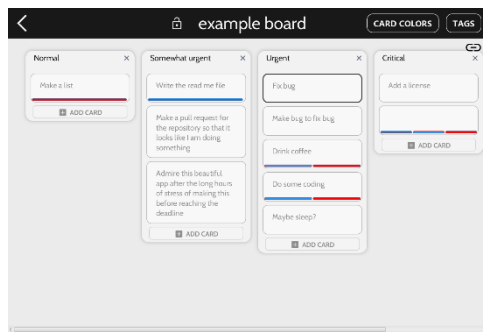
### Rust Software Ray Tracer – (Rust)

A real-time ray tracer written in Rust. This project was made as practice to learn Rust with the aim to know the language a lot better and become a lot more comfortable with it.



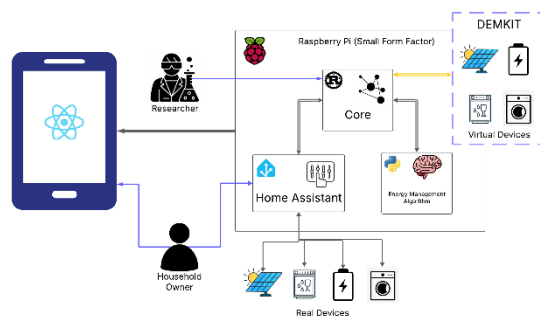
## TALIO – (Java)

Talio is a Java-based planning application. The program is built to let users create, classify and track goals within a simple GUI environment. This project was made for CSE1105 @ TU Delft.



## HHEMS – (Python, Rust) [<https://algtudelft.github.io/HHEMS/>]

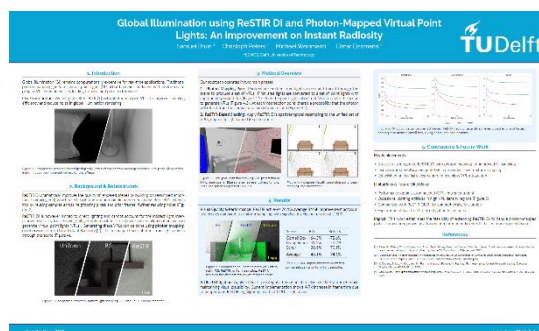
A research-oriented energy management system for optimizing residential energy usage, storage, and generation using forecasting and optimization techniques. Developed at TU Delft.



## BSc Thesis: ReSTIR-based Global Illumination with Photon Mapping – (C++)

[<https://resolver.tudelft.nl/uuid:ddb7ccdd-289b-48a4-bffa-541840f9cfbf/>]

A research project exploring the integration of ReSTIR Direct Illumination with photon-mapped virtual point lights to improve the efficiency and visual quality of global illumination.



## Achievements & Skills

- **Languages:** Rust, C++, Java, Javascript, Typescript, Python, Scala, CUDA
- **Web frameworks:** ReactJS, Svelte, Sveltekit, NextJS, Vue